INCH-POUND

A-A-60085
30 April 2026

COMMERCIAL ITEM DESCRIPTION

TRUCK, AIRCRAFT FUELING-DEFUELING 6,000 GALLONS, A/S32R-11

The General Services Administration has authorized the use of this commercial item description by all federal agencies.

1. SCOPE. This commercial item description (CID) describes the general requirements for a 6,000 gallon capacity aviation refueling truck (herein referred to as the Refueling Vehicle) that will be used for fuel servicing of both military and commercial aircraft.

2. SALIENT CHARACTERISTICS.

2.1 Refueling Vehicle description. The Refueling Vehicle shall be a tank truck that will receive, transport, store, and pump jet fuels in accordance with ASTM D910, ASTM D1655, MIL-DTL-5624, MIL-DTL-25524, MIL-DTL-83133, GOST 10-227, MoD Def Stan 91-091, MoD Def Stan 91-087, ASTM D6615, and ASTM D7566. The Refueling Vehicle shall have a 6,000 gallon capacity (see 2.2.9.1) and shall be capable of the following:

a. Issuing fuel to aircrafts at volumetric flow rates of at least 500 gpm but not more than 600 gpm through a filtration system, a flow meter, a primary and secondary pressure control system, a ground servicing hose, and a single point nozzle.

b. Bottom loading fuel into the cargo tank at flow rates up to 750 gpm with overfill protection.

c. Defueling aircraft into the cargo tank through the ground servicing hose and single point nozzle, the filtration system, and the flow meter at a minimum flow rate of 80 gpm without the assistance of aircraft boost pumps and at flow rates of at least 300 gpm with the assistance of aircraft boost pumps.

Comments, suggestions, or questions on this document should be addressed to WR-ALC/LEEC, 295 Byron Street, Robins AFB, GA 31098-1611. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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2.2 Design and construction. The design shall promote cost effective, life-cycle sustainability by addressing considerations such as incorporating open standards, while satisfying system performance requirements. It shall be designed and constructed so that no parts will work loose in service, and to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service. It shall be weatherproof and designed to prevent the intrusion of water, sand, and dust into critical operating components. It shall also meet all applicable requirements of NFPA 407 for components that are required to be intrinsically safe when used during fueling operations.

2.2.1 Materials, protective coatings, and finish.

2.2.1.1 Protective coatings. Materials that deteriorate when exposed to sunlight, weather, or operational conditions normally encountered during the service life of the item shall not be used or shall have means of protection against such deterioration that does not prevent compliance with the performance requirements specified herein. Protective coatings that chip, crack, or scale with age or extremes of climatic conditions or when exposed to heat shall not be used. Exposed surfaces of fasteners, handles, and fittings shall also be primed and painted. All components that come in contact with jet fuel shall be certified by their manufacturer as being suitable for fuel service and corrosion resistant. See 2.2.4.1 for prohibited hazardous materials.

2.2.1.2 Finish. Unless otherwise specified (see 6.2), the exterior finish color of the Refueling Vehicle and the inner surfaces of compartments shall be semi-gloss green, Color Number 24052 of SAE AMS-STD-595. When specified, the exterior finish color shall be tan 686A, Color Number 33446 of SAE AMS-STD-595. The tank boarding ladder shall not be painted.

2.2.1.3 Rustproofing. The vehicle chassis and cab shall be rustproofed to a tropical level in accordance with FED-STD-297.

2.2.1.4 Exclusion of water. The design of the Refueling Vehicle shall be such as to prevent water leaking into, or being driven into, any part of the Refueling Vehicle interior when either in an operating or travelling configuration. All windows, doors, panels, covers, etc., shall be provided with sealing arrangements such that the entry of water is minimized when these items are correctly closed. Particular care shall be taken to prevent wetting of equipment and heat and sound proofing materials. Sharp corners and recesses shall be avoided so that moisture and solid matter cannot accumulate to initiate localized attack. Sealed floors with suitable drainage shall be provided for storage compartments, engine compartments, and other areas in the Refueling Vehicle that could collect and retain water.

2.2.1.4.1 Fluid traps and faying surfaces. There shall be no fluid traps on the Refueling Vehicle. Faying surfaces of all structural joints, except welded joints, shall be sealed to preclude fluid intrusion.

2.2.1.4.2 Ventilation. Ventilation shall be sufficient to prevent moisture retention and buildup.

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2.2.1.4.3 Drainage. Drain holes shall be provided to prevent collection or entrapment of water or other unwanted fluid in areas where exclusion is impractical. All designs shall include considerations for the prevention of water or fluid entrapment and ensure that drain holes are located to effect maximum drainage of accumulated fluids. The number and location of drain holes shall be sufficient to permit drainage of all fluids when the Refueling Vehicle is on a 10-degree longitudinal slope facing both up and down and on a 10-degree side slope in each direction (right and left side facing up the slope). The minimum size of the drain holes shall be 0.375-inch.

2.2.1.5 Metals. All metal components and piping downstream of the filtration system shall be constructed of aluminum or stainless steel. Zinc coated metals, brass, bronze, or other copper bearing alloys shall not come in direct contact with fuel. Brass components are acceptable in hose couplings/hose reducers and draining valves. Copper components are acceptable in drain lines that render the product as sump fuel.

2.2.1.6 Sense lines. All sense lines (both fuel and air) shall be made of Kynar (a PVDF compound) or equivalent material to provide maximum resistance to UV damage and increase the longevity of the vehicle. Material shall be abrasion and UV resistant as well as meet the operating and storage temperatures specified (see 2.3.1 and 2.3.2). All lines shall have some form of marking or coloring to indicate whether it is an air, fuel, or other type of sense line.

2.2.2 Markings. The Refueling Vehicle shall be marked appropriately using MIL-HDBK-1223 for guidance. Four DOT hazardous material labels, number 1863, shall be mounted on the vehicle, one on each side of the forward 1/3 of the cargo tank, one on the upper right hand corner of the rear of the tank, and one on the front bumper of the truck. The Refueling Vehicle shall also be marked in accordance with 49 CFR 172.302 and National Fire Protection Association (NFPA) 407. All external devices which require an operational or maintenance interface shall be marked in accordance with MIL-STD-130. Markings shall be applied with decals and shall be 1-inch high block letters unless prohibited by the available space. In such cases, the markings shall be the largest size possible, but not less than 0.50-inch high. Markings, Information/Caution shall be Lusterless Black, Color Number 37038 of SAE AMS-STD-595, and Markings, Warning/Danger shall be Lusterless Red, Color Number 31136 of SAE AMS-STD-595.

2.2.2.1 Flammable marking. “FLAMMABLE” in 6-inch letters shall be located on each side and the rear of the cargo tank.

2.2.2.2 No smoking marking. “NO SMOKING WITHIN 50 FEET” in 6-inch letters shall be located on each side and the rear of the cargo tank.

2.2.2.3 Avoid water marking. “CARGO FIRE - AVOID WATER” in 4-inch letters shall be located on each side and the rear of the cargo tank.

2.2.2.4 Emergency tank shutoff marking. “EMERGENCY TANK SHUTOFF” shall be marked in 2-inch letters near each emergency tank shutoff.

2.2.2.5 Ladder marking. “MAX WEIGHT 250 LBS” shall be marked in 2-inch letters on the cargo tank ladder.

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2.2.2.6 Center of gravity marking. The center of gravity of the Refueling Vehicle shall be marked within 1-inch of the calculated center of gravity of the vehicle.

2.2.2.7 Tire inflation pressure markings. Tire inflation pressure shall be marked as near to each tire as possible.

2.2.2.8 Jet fuel marking. Jet fuel product designation shall be marked on each side and the rear of the cargo tank in 6-inch letters. Markings shall be in accordance with the table below unless otherwise noted by the government. Base location will be specified in ordering data (see 6.2).

TABLE I. Product markings.

Refueling Vehicle Final Destination Location	Product Markings
CONUS locations ***except*** Grand Forks AFB ND, Edwards AFB CA, Beale AFB CA	"Jet Fuel Jet A" **and** "F-24"
All OCONUS locations ***and*** Grand Forks AFB ND, Edwards AFB CA, Beale AFB CA	"Jet Fuel JP-8" **and**"F-34

2.2.3 Identification and information plates.

2.2.3.1 Identification plate. An identification plate in accordance with MIL-STD-130 shall be securely attached to the Refueling Vehicle in a readily accessible location. The Refueling Vehicle shall have a Unique Identification (UID) (also known as Item Unique Identification (IUID)) information permanently affixed on or near the respective identification plate(s), marked in accordance with MIL-STD-130. The UID shall be the vehicle identification number (VIN). UID information shall be included as both a 2D Data Matrix (barcode) and human readable markings. The identification plate shall contain the following information:

a. Nomenclature.
b. Air Force vehicle registration number.
c. Part number.
d. Serial number.
e. Date of manufacture.
f. Contract number.
g. Manufacturer's name.
h. Manufacturer's address and phone number.
i. Commercial and Government Entity (CAGE) code.
j. Date of warranty expiration (if applicable).
k. National Stock Number (NSN).
l. Vehicle weight unloaded (pounds).
m. Gross vehicle weight (pounds).

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n. Engine fuel type.

o. Engine oil viscosities for ambient temperatures ranging from -40 to +125 °F.

p. Cargo tank capacity (gallons).

2.2.3.2 Transportation data plate. A transportation data plate shall be securely attached to the Refueling Vehicle in a readily accessible location. The plate shall contain at least the following information:

a. Side and rear silhouette views of the Refueling Vehicle.

b. Horizontal and vertical location of the center of gravity of the Refueling Vehicle in air transportable configuration, marked on the silhouette views.

c. Shipping weight.

d. Loading cubage.

e. Overall height, width, and length.

f. Front and rear axle loads.

g. Tie down information.

2.2.3.3 Lubrication data plate. The Refueling Vehicle shall be provided with a permanently-marked lubrication data plate constructed of a non-corroding metal, mounted on the exterior surface of the Refueling Vehicle that shall direct attention to all lubrication fittings and components that require lubrication. The plate shall identify the type and grade of lubricant required for operational temperatures specified in 2.3.1.

2.2.4 Environment, Safety, and Occupational Health (ESOH).

2.2.4.1 Hazardous material. The design shall minimize and control hazards associated with the inclusion or use of hazardous or toxic materials and the generation of toxic or noxious gases. The Refueling Vehicle shall not generate or use Class I or Class II Ozone Depleting Substances (ODS) during operation, maintenance, or disposal. Class I ODS and hazardous materials shall not be used in any system, component, or process. The Refueling Vehicle shall not contain or use either hexavalent chromium or cadmium without written approval by the procuring activity. Hazardous materials are defined in AFI 32-7086; Class I and Class II ODS are defined in 40CFR82.

2.2.4.2 Component protection. All space in which work is performed during operation, service, and maintenance shall be free of hazardous protrusions, sharp edges, or other features which may cause injury to personnel. All rotating and reciprocating parts and all parts subject to high operational temperatures or subject to being electrically energized, that are of such nature or so located as to be hazardous to personnel, shall be guarded or insulated to eliminate the hazard. All wires, cables, tubes, and hoses shall be supported and protected to minimize chafing and abrasion and shall be located so as to provide adequate clearance from moving parts and high operational temperatures. Grommets shall be provided wherever wires, cables, tubes, or hoses pass through bulkheads, partitions, or structural members.

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2.2.4.3 Foreign object damage (FOD). All loose metal parts, such as pins or connector covers, shall be securely attached to the Refueling Vehicle with wire ropes or chains. "Dog tag" style beaded chains shall not be provided. Removable panels, if provided, shall be attached with captive fasteners. Tire valve stem caps shall be made of plastic.

2.2.4.4 Fire extinguisher. Two dry chemical fire extinguishers, Type 1, Class 2, Size 20, in accordance with A-A-393, shall be mounted with one on either side of the truck in a position that is readily accessible to personnel standing on the ground. The fire extinguishers shall be located so that they are protected from tire splash.

2.2.4.5 Noise. The design shall ensure that noise created by the Refueling Vehicle is compatible with the environment and minimize exposure of personnel to noise hazards during operations and maintenance activities. Maximum noise levels during operation shall not exceed 85 dBA.

2.2.4.6 Electrostatic discharge (ESD). The design of the Refueling Vehicle shall preclude equipment damage due to ESD, protect personnel from electrical shock due to static charging, and prevent ignition of explosive atmospheres due to sparking.

2.2.4.7 Grounding cables and reels. Two Type I, spring rewind, grounding reels with cables and cable guides shall be installed side by side on the driver's side of the truck. The reels and cables shall be in accordance with A-A-50696. One cable shall have a welder style clamp, and the other cable shall have a grounding plug. The cables shall have cable stops to prevent the clip or plug from striking the cable guide. The cables shall be electrically bonded to the tanker chassis frame. Each reel shall have 75-ft of cable.

2.2.5 Electromagnetic interference (EMI). The Refueling Vehicle shall be in accordance with the following radiated emission and susceptibility requirements of MIL-STD-461: RE102 (to the limits of FIGURE RE102-4, Navy Fixed & Air Force curve) and RS103 (to the levels of TABLE VII) for the following ranges:

a. 30 MHz - 1 GHz at 10 Volts per meter (V/m).

b. 1 GHz - 18 GHz at 50 Volts per meter (V/m).

2.2.6 Human systems integration. The Refueling Vehicle shall be designed in accordance with MIL-STD-1472 for ease of operation, inspection, and maintenance, including the use of arctic mittens and Mission-Oriented Protective Posture (MOPP) Level 4 Chemical Warfare Gear. Chemical Warfare Gear is not required for preventive maintenance or major corrective maintenance.

2.2.7 Fastening devices. All screws, bolts, nuts, pins, and other fastening devices shall be properly designed, manufactured, and installed with adequate means of preventing loss of torque or adjustment. Cotter pins, lock washers, or nylon patches shall not be used for this purpose, except for the attachment of trim items or as provided in commercial (see 6.3.4) components. Tapped threads shall have a minimum thread engagement in accordance with Table I.

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TABLE II. Minimum thread engagement.

Material	Minimum Thread Engagement
Steel	1.0 times the nominal fastener diameter
Cast iron, brass, or bronze	1.5 times the nominal fastener diameter
Aluminum, zinc, or plastic	2.0 times the nominal fastener diameter

2.2.8 Welders and welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface parts to be welded shall be free from rust, scale, paint, grease, and other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the welded parts. Welds shall transmit stress without cracking or permanent distortion when the parts connected by the welds are subjected to test, proof, and service loadings.

2.2.9 Cargo tank and related components.

2.2.9.1 Cargo tank. The Refueling Vehicle shall have a single-compartment cargo tank in accordance with all requirements of NFPA 385, NFPA 407, and 49 CFR 178 (DOT 406). The cargo tank shall have a capacity of 6,000 gallons plus a minimum of three percent expansion space. The cargo tank shall be constructed of aluminum. There shall be no metal-to-metal contact between the tank mounting brackets and the truck chassis mounting brackets.

2.2.9.2 Baffles. Baffles should be spaced approximately equally throughout the cargo tank. The baffles shall be designed so that personnel can access the entire tank. To prevent static electricity generation caused by free falling fuel, the baffles shall be designed to allow uniform filling of the tank through the bottom loading system at flow rates up to 750 gpm (see 2.2.9.11).

2.2.9.3 Manhole. The cargo tank shall have a minimum 20-inch diameter manhole. The manhole shall prevent standing water from entering the cargo tank.

2.2.9.4 Sump. The bottom of the cargo tank shall have an unobstructed path sloped towards a minimum 12 gallon sump.

2.2.9.5 Sump drain. The sump shall have a minimum 1.5-inch minimum self-closing drain valve. The drain valve shall be directed toward the ground, guarded from damage, accessible without crawling under the tank, and designed to prevent fuel contact with the operator.

2.2.9.6 Cleaning stubs. In the top of each section created by the baffles (except for the section containing the manhole) a minimum three inch diameter cleaning stub shall be installed. All stubs shall have caps.

2.2.9.7 Vent system. The cargo tank vent system shall be a two part system. The primary system shall be an automatic vent system. The automatic vent system shall be interlocked and fail-safe during refueling, defueling, and bottom loading operations so that flow cannot commence unless the vent(s) are fully open. The automatic vent system shall be sized for a cargo tank fill rate of 750 and a single-point nozzle discharge rate of 600 gpm. The secondary system shall be an emergency vent in accordance with 49 CFR 178 (DOT 406).

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2.2.9.8 Catwalk. A 30-inch wide catwalk shall span the full length of the top of the cargo tank and the manhole and include a landing platform for the rear-tank boarding ladder .The catwalk shall support a load of 300 pounds per square foot without permanent deformation and shall have a drainable surface. The catwalk shall have scuff guards on both sides and shall be equipped with fall protection in accordance with 29 CFR 1910.

2.2.9.9 Cargo tank ladder. A cargo tank boarding ladder capable of supporting a single rung load of 300 pounds shall provide access to the catwalk from the rear center of the cargo tank. A fixed step below the rear bumper shall facilitate access to the ladder.

2.2.9.10 Bottom loading system. The Refueling Vehicle shall be equipped with an electronically controlled, air-powered, automatic shut-off bottom loading system. The system shall be capable of filling the cargo tank from a pressurized fuel source through a single point receptacle with the engine off. The bottom loading system shall accept fuel flow up to 750 gpm.

2.2.9.11 Inlet receptacle. The bottom loading inlet receptacle shall be located on the driver's side of the cargo tank. The inlet shall include a single-point nozzle adapter and dust cover in accordance with MIL-A-25896 and MS24484-2.

2.2.9.12 Inlet manifold. The bottom loading inlet manifold shall connect the bottom loading inlet receptacle and the defuel tank fill pipe to the cargo tank.

2.2.9.13 Bottom loading shutoff valve. A manual bottom loading shutoff valve shall be installed in the bottom loading inlet manifold. The valve shall have no less than eight segmented and latched positions from fully closed to fully open.

2.2.9.14 Bottom loading internal valve. An air-operated bottom loading internal valve shall be interlocked so that it only opens after the truck parking brakes are set and the automatic vent system is open. The bottom loading internal valve shall automatically activate when a single point nozzle is connected to the bottom loading inlet receptacle. The valve must be equipped with a fusible link or a plug that closes the valve in the event of a fire.

2.2.9.15 Emergency tank shutoff valve. The emergency tank valve shall be located between the tank and the in-line strainer. The emergency tank valve shall be equipped with a fusible link or plug that closes the valve in the event of a fire.

2.2.9.16 Overfill prevention. A dual, independent, automatic, electronically controlled, high level shut off system shall be installed to stop fuel flow during bottom loading and defueling operations when the cargo tank is full. Electrical power to the high-level shutoff system shall be supplied when the Refueling Vehicle is in Defuel mode or Bottom loading mode. The system used shall be compatible with Scully ST-15 single point overfill prevention control unit, Scully Intelliltrol 2, or an equivalent system that uses a dual socket adapter and includes a pretest momentary-on push button located on the bottom loader control panel for test verification of the secondary tank-level sensor. The socket adapter shall be equipped with ground verification.

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2.2.9.17 Tank level sensors. The cargo tank shall include independent primary (HI) and secondary (HI-HI) tank level sensors to prevent overfill for both bottom loading and defueling operations. The primary tank level sensor shall be installed approximately 0.75-inches lower than the secondary tank level sensor. The signals through the primary and secondary sensors shall be continually monitored and any fault in the electronic circuits shall result in a shutdown of any defueling or bottom loading operations. If the primary tank level sensor (HI) has contact with fuel, the primary pressure control system (see 2.2.10.11.1) shall deactivate the deadman and terminate fuel flow in any of the modes mentioned above. If the secondary tank level sensor (HI-HI) has contact with fuel, air pressure shall cause the bottom loading internal valve to close within four seconds.

2.2.9.18 High level shutoff override. The high level shutoff system shall include an override in order to allow gravity discharge of fuel if the secondary sensor (HI-HI) is wet. The override shall be located near the bottom loading inlet and implemented by a momentary-on push button protected with a tamper proof cover that can be secured with a padlock. The padlock will be installed by the government upon receipt. Release of the override button shall close the bottom loader internal valve within four seconds.

2.2.10 Pumping system.

2.2.10.1 Operation selection. The pumping system includes the piping, tank suction header, pump, dead man control, filtration system, flow and pressure controls, flow meter, operator controls, automatic valves, hose reels, and hoses with nozzles. The Refueling Vehicle shall allow the operator to perform one of the following operations:

a. Fuel servicing - high pressure.

b. Fuel servicing - low pressure.

c. Defuel.

d. Recirculation.

e. Bottom loading.

f. Overwing fuel servicing.

2.2.10.2 Piping. The piping shall be seamless schedule 40 aluminum in accordance with ASTM B241/B241M or seamless schedule 10 TP 304L stainless steel in accordance with ASTM A312/A312M. Pipe and fittings shall be either flange or groove connected, or a combination of both. All piping shall include provisions for the relief of fuel pressure and hoses after a pumping operation is completed.

2.2.10.3 Tank suction header. The tank suction header shall include an air operated internal tank suction valve (the “emergency” valve), a check valve, a suction maintenance shutoff valve, and an in-line strainer. The suction maintenance shutoff valve shall be a manual quarter turn design that can be operated while standing or kneeling on the left or right side of the truck and mounted prior to and in close proximity to the in-line strainer.

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2.2.10.4 Emergency tank shutoff controls. One manually operated emergency tank shutoff control shall be located on each side of the cargo tank in close proximity to the emergency tank valve (see 2.2.10.3). When activated, the emergency tank shutoff controls shall cut off air pressure to the emergency tank valve thereby causing is to immediately close and terminate fuel flow from the tank regardless of whether the dead man device is activated or deactivated. The emergency tank shutoff controls shall not be recessed and shall be of the toggle switch or push-pull plunger type. If a push-pull plunger switch is used the switch shall have a two-position detent and shut off fuel flow when it is pushed in. The emergency valve shall be equipped with a fusible link or plug that closes the valve in the event of a fire.

2.2.10.5 Strainer. An in-line Y-style strainer shall be installed in the piping between the tank suction header and the pump. The strainer shall have a serviceable 10 to 20 mesh screen that is rated for flows of 600 gpm. The strainer shall be positioned in an easily accessible location for servicing.

2.2.10.6 Pump. The pump shall be a self-priming, centrifugal, aviation fuel dispensing pump. The pump shall be driven either directly or hydraulically by the truck transmission PTO or by a split shaft gear box (see 2.8.1.1). A drain plug shall be installed at the pump housing low point. A bypass system or a relief valve shall be provided to protect the pump.

2.2.10.7 Pump discharge shutoff valve. A pump discharge shutoff valve shall be located both upstream and downstream of the pump. There shall be no devices, branch connections, or bypass/relief connections between the shutoff valves and the pump. The valves shall be a manual, quarter turn design and shall be operable while standing or kneeling on the left or right side of the truck.

2.2.10.8 Dead man control. An electric or pneumatic dead man controller shall be provided to start and stop flow for any operation that is selected on the control panel. The dead man shall be a robust design suitable for refueler service with either a squeeze handle or push-button device. Upon release of the dead man, any fuel flow shall terminate within 3 seconds. The dead man cable or hose shall have a length of 80-ft. If an electric dead man is used, a cable reel with spring rewind shall be provided. If a pneumatic dead man is used, a powered cable reel with a momentary-on push button shall be used for rewinding the hose. In this case, the cable reel shall incorporate the use of a fairlead to guide the cable. With both the spring rewind and powered cable reels, a hose/cable stop shall be installed to prevent the dead man device from striking the reel. The cable reel shall be conveniently located near the operator control panel.

2.2.10.9 Flow meter. All fuel flow through the pumping system shall be metered through a flow meter. The meter shall be a positive displacement meter that is certified to be accurate to ±0.2 percent between 50 and 600 gpm in accordance with NIST Handbook 44. The meter shall be capable of registering flow down to 2.5 gpm. The flow meter shall have a resettable totalizer capable of displaying the amount of fuel delivered in gallons.

2.2.10.10 Automatic valves. All valves that are required to implement the different operations (see 2.2.10.1) shall be mechanically controlled. The valves will be open/closed based on which operating mode is selected on the control panel (see 2.2.12.2). The operator shall not have to manually open or close any valves to perform any of the operating modes. All valves shall be intrinsically safe.

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2.2.10.11 Pressure control system. The pressure control system shall consist of an independent primary and secondary pressure control system. The pressure control system shall limit the single point nozzle pressure to a maximum peak surge pressure of 120 psi during a complete aircraft fueling valve(s) closure occurring in a maximum timeframe of two seconds. Any surge pressure developed shall not exceed the pressure rating of any single component in the Refueling Vehicle pumping system. The pressure control system shall include a pump bypass system (see section 2.2.10.6). The pressure control system shall sense the single point nozzle pressure through a pressure sensing type component that consistently and repeatedly simulates the fuel pressure at the single point nozzle. Locked in pressure at the single point nozzle shall not exceed 55 psi for more than 15 seconds after flow stops. The system shall resume from an induced no-flow condition to primary control system controlled flow within 15 seconds by disengaging and then re-engaging the dead man.

2.2.10.11.1 Primary pressure control system. The primary pressure control system shall be set to control and limit the single-point nozzle pressure to the maximum pressure setting associated with each servicing mode (see section 2.7). The primary pressure control system shall not automatically vary the Refueling Vehicle's engine speed. When the primary pressure control system is activated, fuel flow shall continue.

2.2.10.11.2 Secondary pressure control system. The secondary pressure control system shall be set to control and limit the single-point nozzle pressure to a maximum of 55 psig if the primary pressure control system should fail. The secondary pressure control system shall not automatically vary the Refueling Vehicle's engine speed. When the secondary pressure control system is activated, the Refueling Vehicle shall not shut off. If a pressure switch is used as a secondary control, it shall not be electric.

2.2.10.12 Pumping compartment. An enclosed pumping compartment shall be located between the cab and the cargo tank. The entire pumping system including the associated piping, filtration system, flow meter, and any hose reels shall be housed within this compartment. However, the pump shall not be housed within this compartment.

2.2.10.12.1 Pumping compartment design. The pumping compartment shall consist of a structural frame, a floor, single panel walls, and structural roof panels that form the compartment into a weather resistant enclosure. The compartment shall have natural air circulation that prevents the accumulation of fuel vapors. The roof panel shall have a continuous longitudinal center hinge and shall support a load of 300 pounds per square foot without permanent deformation. The roof panels shall be securely latched to the structure so that they may remain secure at travel speeds up to 65 mph. The roof panel shall also include lifting points to aid in removal.

2.2.10.12.2 Pumping compartment doors. A roll-up door shall be provided on each side of the pumping compartment. Each door shall open enough so that maximum a maximum door opening is obtained. The doors shall be capable of being latched in the open position. If the open door exceeds six feet above the ground when in the open position, an assist strap shall be included to aid in closing. Latch handles shall extend the full length of the doors and be capable of release from any point while wearing both arctic mittens and MOPP level 4 chemical warfare gear. The driver side door shall be sufficiently wide to provide access to the pumping system controls, the flow meter, and any hoses. The passenger side door shall be sufficiently wide for inspection and service access to any of the components located within the compartment.

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2.2.11 Filtration.

2.2.11.1 Filtration system. The filtration system shall be rated for 600 gpm and 150 psig. The filter vessel and elements shall be qualified in accordance with the requirements set forth in the most recent edition and revision of EI-1581 Specification and Qualification Procedures for Aviation Jet Fuel Filter Separators, Category M, Type S, Two Stage system configuration.

2.2.11.2 Filter vessel. The filter vessel shall be designed and constructed in accordance with the most recent edition and revision of ASME Boiler and Pressure Vessel Code Section VIII and EI 1596 Design and Construction of Aviation Fuel Filter Vessels. The vessel shall be inspected and stamped by a qualified ASME inspector. The filter vessel shall be constructed of aluminum and shall include a ball valve vent to aid in draining of the filter vessel. The ball valve vent shall include a check valve for only allowing air in. The filter vessel shall remain full regardless of the level of fuel in the cargo tank during pumping operation or while taking a sample from the drain. The filter vessel drain shall have a self-closing valve, accessible by a person standing on the ground.

2.2.11.3 Water slug shutoff device. A water-slug shutoff device that uses a float-actuated valve or an electronic sensor shall be installed in the filter vessel. The device shall stop fuel flow within 10 seconds when a predetermined water level is reached inside the vessel. Fuel flow shall resume within 30 seconds when the water level is lowered to the predetermined level. If a float-actuated valve is used, the water-slug shutoff device shall provide for manual movement of the float through an external testing mechanism that is readily accessible to a person standing on the ground. If an electronic sensor is used, it shall have a light or other visual means showing proper working order. The electronic sensor shall be in compliance with EI 1592.

2.2.11.4 Filter vessel water drain. The water sump on the filter vessel shall be equipped with a one inch manually operable ball-type valve with a spring return handle that renders the valve normally closed. The drain valve shall be directed toward the ground, guarded from damage, accessible without crawling under the truck, and designed to prevent fuel contact with the operator.

2.2.11.5 Element sealing. All filter vessel element connections shall be sealed in accordance with EI 1581.

2.2.11.6 Differential pressure gauge. A non-calibrating differential pressure gauge shall be installed to continuously monitor the pressure drop across the filter coalesce stage during operation. The gauge shall register and hold the highest pressure drop achieved during operation. The gauge shall be located so that an operator can see and access it while standing at the control panel.

2.2.11.7 Fuel sampling. A fuel sampling valve shall be provided downstream of the filtration system and prior to the flow meter. The valve shall be easily accessible while standing at the operator control panel. A Gammon MiniMonitor tray (model GTP-9589) shall also be installed downstream of the filtration system and prior to the flow meter. The tray will allow the Air Force the capability to connect a Gammon MiniMonitor (model GTP-9571) in-line, closed-loop, solids and water sampler.

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2.2.12 Controls.

2.2.12.1 Control panel. The control panel shall be located on the driver's side of the truck in a location that is easily accessible to a person standing on the ground. The control panel shall be easily accessible for maintenance from the rear and may swing out if necessary. The control panel will contain pumping system controls that are grouped together by function. Instructions and diagram plates shall be mounted on or adjacent to the panel. Once a pumping mode is selected, manipulation of controls shall not be required. The only exception to this is the operation of the dead man or the operation of the auxiliary throttle. The vent system shall be automatically activated by the control system when the operator selects a pumping mode. It should be noted that the pumping system will not operate with the vents closed (see 2.2.9.8).

2.2.12.2 Pumping mode selection. The control panel shall have a selector switch or switches with on-off indicator lights that allow the operator to select one of the following operation modes:

a. Fuel servicing - high pressure.

b. Fuel servicing - low pressure.

c. Defuel.

d. Recirculation.

e. Bottom loading.

f. Overwing fuel servicing.

g. Off.

2.2.12.3 Flow meter register. A flow meter register shall be installed on the control panel. It shall have a LED digital display that is readable from a distance of 15-feet during daytime or nighttime operations. The register shall display flow rate in gpm and the totalized flow in gallons. It shall have both a resettable counter and a non-resettable totalizer.

2.2.12.4 Tachometer. A tachometer shall be installed on the control panel. It shall have a LED digital display that is readable from a distance of 15-feet during daytime or nighttime operations.

2.2.12.5 Throttle control. A throttle controller shall be installed on the control panel. The controller shall allow the operator to manually control the engine speed during fueling, defueling, or recirculation operations.

2.2.12.6 Fuel pressure gauges. Gauges displaying pump suction pressure, discharge pressure, and single point nozzle pressure shall be installed on the control panel. The gauges shall have LED digital displays that are readable from a distance of 15-feet during daytime or nighttime operations. Pressure readings shall be taken from pressure transmitters that are incorporated in the flow and pressure control system.

2.2.12.7 Air pressure gauge. A mechanical air pressure gauge shall be installed on the control panel to display available truck air pressure.

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2.2.12.8 Emergency engine shutoff. An emergency engine shutoff shall be installed on the control panel. The shutoff shall be a red mushroom style push-pull type button that is at least one inch in diameter. The shutoff shall shut down the truck engine without the use of any other control.

2.2.12.9 Cargo tank high level pretest. A cargo tank high-level pretest button shall be installed on the control panel. The button shall be provided to test the overfill prevention system (see 2.2.9.16). Indicator lights (see 2.2.12.10) shall be provided to indicate whether the high-level sensor is wet or dry. Upon activation of the high-level pretest button, the flow fuel shall terminate to simulate a fully loaded cargo tank.

2.2.12.10 Cargo tank high level indicator lights. Two high level indicator lights shall be located on the control panel. The first light shall be a "Dry" indicator light that illuminates green. When this light is illuminated, the primary tank level sensor is not in contact with fuel thus indicating that the cargo tank is not yet at full capacity. The second light shall be a "Wet" indicator light that illuminates red. When this light is illuminated, the primary tank level sensor is in contact with fuel thus indicating that the cargo tank is at full capacity. When the high level pretest button (see 2.2.12.9) is pushed the "Wet" indicator light shall illuminate.

2.2.12.11 Lamp test button. A momentary-on push button shall be installed on the control panel to test the control panel lights.

2.2.13 Hoses assemblies.

2.2.13.1 Ground servicing hose assembly. The ground servicing hose shall consist of 75-feet of 2.5 inch EI 1529, Grade 2, Type C hose. The hose shall have a ball valve style strainer (Cla-Val P/N E349GF, JC Carter P/N 64015, or equivalent). If ordered for extreme climatic temperatures (see section 6.2) the hose shall consist of 75-feet of EI 1529, Grade 2, Type CT hose. This hose shall also have a ball valve style strainer (Cla-Val P/N E349GF, JC Carter P/N 64015, or equivalent).

2.2.13.2 Ground servicing nozzle. The ground servicing nozzle for either hose assembly shall be D-3 universal single point nozzle. The nozzle shall have a 60 mesh stainless steel strainer and an automatic vacuum breaker.

2.2.13.3 Ground servicing hose reel. The ground servicing hose reel shall be an air powered or electric powered commercial hose reel installed on the driver's side of the truck. The reel shall have a clutch released drag brake, a manual rewind, and a locking mechanism that will ensure the hose does not come loose during transportation. The locking mechanism shall be easily accessible by an operator standing on the ground at the control panel. The hose reel shall be capable of holding 75 working feet (from the hose roller guide to the end of the single point nozzle) of 2.5-inch non-collapsible hoses with 2.5-inch National Pipe Threat Tapered Thread (NPT) male threads on each end. A momentary-on push button for rewinding the hose shall be provided in an easily accessible location adjacent to the reel. Unless the hose reel is of the single wrap type, stainless steel hose guide rollers shall be used. The rollers shall have permanently lubricated metal bearings. A manual isolation valve shall be provided for the hose reel.

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2.2.13.4 Overwing hose. The overwing hose shall consist of 75-feet of 1.5-inch EI 1529, Grade 2, Type C hose.

2.2.13.5 Overwing hose nozzle. The overwing nozzle shall be an OPW model 295AF nozzle or equivalent. The overwing nozzle shall have a 60 mesh stainless steel strainer and the grounding wires shall be jacketed.

2.2.13.6 Overwing hose reel. The overwing hose reel shall be an air powered or electric powered commercial hose reel installed on the driver's side of the truck. An overwing hose isolation valve shall be provided. The hose reel shall be capable of holding 75 working feet (from the hose roller guide to the end of the overwing nozzle) of 1.5-inch non-collapsible hose. The reel shall have an adjustable released drag brake that allows for a slow and controlled deployment of the overwing hose. The reel shall have features to prevent over-travel during deployment and rewind. Unless the hose reel is of the single wrap type, stainless steel hose guide rollers shall be used. The rollers shall have permanently lubricated metal bearings. A momentary-on push button for rewinding the hose shall be provided in an easily accessible location adjacent to the reel. The button shall be conveniently located so that the operator can guide the overwing hose with one hand and operate the rewind button with the other hand.

2.2.14 Cab and chassis.

2.2.14.1 Cab. The Refueling Vehicle shall have a two door cab with a two person capacity. The cab shall be equipped with two individual air ride seats. All necessary controls, instruments, gauges, and driver's aids required to operate the truck shall be located in readily accessible locations inside the cab. The cab shall have air conditioning.

2.2.14.2 Mirrors. The Refueling Vehicle shall be equipped with both left and right side mirrors. The mirrors shall be electronically adjustable from the driver's position within the cab. The mirrors shall be large enough to provide a clear and distinct view of the entire length of the truck and rear bumper.

2.2.14.3 Backup aids. The Refueling Vehicle shall be equipped with backup lights and a backup alarm. The alarm shall meet the Type D (87dbA) requirements of SAE J994.

2.2.14.4 Equipment and accessories. In addition to the cab requirements of FED-STD-807 for equipment and accessories, the Refueling Vehicle shall be equipped with an hour meter and two switch operated reading lights for use while seated or while standing at either door during nighttime operations.

2.2.14.5 Chassis. The Refueling Vehicle chassis shall be a standard commercial chassis that is in accordance with FED-STD-807. The chassis shall be a Type 1, Class F, six by four chassis with a two door cab. The pumping system for this vehicle shall be mounted to the chassis. The Refueling Vehicle shall be towable from the front with or without the front tires off the ground.

2.2.14.6 Lighting. Unless lights are provided as standard equipment for the commercial chassis, all lighting for the Refueling Vehicle shall be LED with the exception of headlights. In addition to the vehicle lights and reflectors required by the Federal Motor Carrier Safety Regulations (49 CFR 393), the lights described in the following subsections shall also be provided.

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2.2.14.6.1 Pumping system lights. The pumping system components and controls required for operation during nighttime servicing shall be illuminated to a minimum of 50-foot candles. Light switches for these lights shall be installed on the control panel (see 2.2.12.1).

2.2.14.6.2 Night servicing lights. Two adjustable, vapor proof, service lights shall be installed to facilitate night timer servicing. One light shall be installed at the rear of the cab on the driver's side. The other light shall be installed at the rear of the cargo tank on the driver's side. The lights shall be approximately eight feet above the ground. Each light shall have its own on/off switch and be adjustable by a control handle that is within reach of a person standing on the ground. The combined light levels of these two lights shall illuminate the entire side of the vehicle.

2.2.14.7 Clean-up materials compartment. A six cubic foot, weather resistant compartment shall be mounted on the passenger side frame rail, below the cargo tank. The compartment will be used for storage of 200 pounds of environmental clean-up materials. The compartment shall have a hinged, hatch-type cover with hold-open features.

2.2.14.8 Overwing hose assembly compartment. The overwing hose assembly shall be installed in a compartment on the driver's side of the truck. The manual overwing hose isolation valve shall be located in this compartment. The compartment door shall swing down to open in order to allow unobstructed access for the operator during use. The latch handle shall be operable while wearing arctic mittens and MOPP Level 4 Chemical Warfare Gear and releasable from any point.

2.2.14.9 Fenders. The Refueling Vehicle shall have fenders or similar protective covers over the rear wheels.

2.2.14.10 Mud flaps. Mud flaps shall be installed at the rear of each fender in accordance with SAE J682.

2.3 Environmental conditions.

2.3.1 Operating temperature range. The Refueling Vehicle shall be capable of operating in ambient temperatures ranging from -40 to 125 °F.

2.3.2 Storage temperature range. The Refueling Vehicle shall be capable of being stored in ambient temperatures ranging from -40 to 125 °F.

2.3.3 Precipitation.

2.3.3.1 Rain. The Refueling Vehicle shall be capable of storage and operation during rainfall of 5-inches per hour for three consecutive hours and 10-inches per hour for 10 consecutive minutes, with winds of up to 35 knots; and with 6-inches of rain per hour impinging on the Refueling Vehicle at angles from vertical to 45 degrees for 30 consecutive minutes.

2.3.3.2 Snow. The Refueling Vehicle shall be capable of storage and operation during accretion of wet snow up to 2-inches per hour for at least 12 hours.

2.3.3.3 Ice. The Refueling Vehicle shall be capable of storage and operation with ice accretion up to 1.5-inches on exposed horizontal surfaces. An operator may use an ice scraper for five minutes during the start-up process.

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2.3.3.4 Humidity. The Refueling Vehicle shall be capable of storage and operation with relative humidity ranging from 0 to 100 percent.

2.3.3.5 Solar radiation. The Refueling Vehicle shall not be adversely affected by full time exposure to solar radiation, such as those conditions encountered in desert environments.

2.3.3.6 Fungus. All materials used in the Refueling Vehicle shall be fungus resistant or shall be suitably treated to resist fungus. Materials treated for fungus resistance shall retain their original electronic and physical properties, shall not present toxic hazards, and treatment shall last for the entire service life of the part. The Refueling Vehicle shall be suitable for operation and storage in conditions encountered in a tropical environment.

2.3.3.7 Salt fog. The Refueling Vehicle shall be capable of storage and operation in high temperature, high humidity, salt laden, seacoast environments without damage or deterioration of performance.

2.3.3.8 Sand and dust. The Refueling Vehicle shall be capable of storage and operation during exposure to wind-blown sand or dust without damage or deterioration of performance.

2.4 Weight and dimensions. Overall weight and dimensions in air transport configuration (see 2.5.2) shall not exceed:

Weight	30,000 pounds.
Length	460 inches.
Width	106 inches.
Height	102 inches.

2.5 Transportability.

2.5.1 Surface transportability. The Refueling Vehicle shall be transportable via all modes of surface shipment (highway, rail, and water) in accordance with MIL-STD-1366 and shall be capable of withstanding the mechanical shock and vibration characteristics of highway, rail, and water transport, except that design for rail impact testing (see 5.2.5 of MIL-STD-1366) is not required. To prevent damage, the Refueling Vehicle shall not be lifted by crane/slings. The Refueling Vehicle shall only be driven on and off the mode of transportation used for shipping.

2.5.2 Air transportability. The Refueling Vehicle shall be transportable on C-130, C-17, and C-5 aircraft in accordance with MIL-STD-1791. Air transport configuration for the Refueling Vehicle shall be with the cargo tank and pumping systems empty. In all air transport configurations, the Refueling Vehicle shall be capable of being restrained and withstanding, without loss of serviceability, 2.0 G up and 4.5 G down accelerations, and shall be capable of being restrained and withstanding, without loss of structural integrity, 3.0 G forward, 1.5 G aft, and 1.5 G lateral accelerations. The Refueling Vehicle shall be equipped with pressure relief devices or configured for air transport to prevent any part from becoming a projectile in the event of catastrophic loss of aircraft cabin pressure. The Refueling Vehicle shall drive on and off the aircraft.

2.5.2.1 Shoring. If shoring is necessary for loading the Refueling Vehicle on a C-130, C-17, or C-5 aircraft, the shoring shall be supplied with each Refueling Vehicle.

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2.5.2.2 Axle weight. Axle weight shall not exceed 13,000 pounds.

2.5.2.3 Tire pressure. Tire pressure shall not exceed 100 pounds per square inch (psi) and shall not be reduced for air transport.

2.5.2.4 Equipment removal and reconfiguration. Preparation for air transport shall take no more than 15 minutes and restoration to operating configuration shall take no more than 15 minutes for 2 persons using common hand tools (see 6.3.5). All equipment removed shall be stored on the Refueling Vehicle; caps and plugs shall permit moving and storage in transport configuration.

2.5.3 Tie downs. The Refueling Vehicle shall be symmetrically restrained during air and ground transport. Tie down points shall be rated at a minimum of 25,000 pounds, marked for capacity, with a clear opening compatible with MIL-DTL-25959 and MIL-PRF-27260 tie down devices. Each end of each tie down device shall terminate at a tie down point and not pass through any other tie down point. There shall be no interference between tie down devices and the Refueling Vehicle. The tie down provisions shall be in accordance with 4.1 through 4.12 of MIL-STD-209. The axles may be used as tie down points. However, no more that 50 percent of the restraint in any direction may be provided by the axles.

2.6 Maintainability. The Refueling Vehicle shall be designed for maintainability in accordance with 5.9 through 5.9.18 of MIL-STD-1472; forces shall not exceed those specified for both males and females.

2.6.1 Inspection and servicing provisions.

a. Pre-use inspections and servicing tasks shall not require tools.

b. Routine service and preventive maintenance shall not require special tools (see 6.3.10).

c. Drain plugs and filters shall be directly accessible from the ground and oriented to have unimpeded drainage to a catch pan.

d. The Refueling Vehicle shall be designed with maximum usage of sealed lifetime lubrication bearings.

e. The Refueling Vehicle shall be designed so the correct oil and coolant levels can be visually checked while the unit is running.

2.6.2 Special tools. The design of the item shall minimize the requirement for special tools (see 6.3.10). All special tools shall be provided with, and stored on, the Refueling Vehicle.

2.6.3 Diagnostic software. A copy of any diagnostic software required or recommended for maintaining the Refueling Vehicle shall be provided with each Refueling Vehicle on CD-ROM or DVD-ROM.

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2.7 Performance.

2.7.1 Fuel servicing - high pressure mode. When operating in high pressure mode, the throttle controller will be used to control flow rate and nozzle pressure until the primary pressure control system limits the pressure to no more than 50 ±5 psi. There shall also be an independent secondary pressure control system that will limit the nozzle pressure to 55 psi. In the event that the secondary pressure control system is activated, the Refueling Vehicle shall not shut off. When operating in this mode, the pumping system shall be capable of issuing fuel to an aircraft from the cargo tank, through the filtration system, the flow meter, ground servicing hose, and the single point nozzle at a flow rate of at least 500 gpm but not more than 600 gpm.

2.7.2 Fuel servicing - low pressure mode. When operating in low pressure mode, the throttle controller will be used to control flow rate and nozzle pressure until the primary pressure control system limits the pressure to no more than 30 ±5 psi. There shall also be an independent secondary pressure control system that will limit the nozzle pressure to 55 psi. In the event that the secondary pressure control system is activated, the Refueling Vehicle shall not shut off. When operating in this mode, the pumping system shall be capable of issuing fuel to an aircraft from the cargo tank, through the filtration system, the flow meter, ground servicing hose, and the single point nozzle at a flow rate of at least 300 gpm but not more than 600 gpm.

2.7.3 Defuel mode. With the fuel level height at 10-feet above the ground, the pumping system shall be capable of defueling fuel from an aircraft through the single-point nozzle and hose, the filtration system, the flow meter, and into the cargo tank at a flow rate of at least 80 gpm without the assistance of aircraft boost pumps and at least 300 gpm with the assistance of aircraft boost pumps. With the fuel level at least three feet below ground level, the pumping system shall be capable of defueling fuel from bowser/drop tank through a 15-foot long, 1.5-inch pigtail hose connected to the single-point nozzle, the filtration system, the flow meter, and into the cargo tank at a flow rate of at least 50 gpm. Defueling operations shall terminate upon fuel contact with the primary tank-level sensor thus terminating flow.

2.7.4 Recirculation mode. In recirculation mode, the pumping system shall recirculate fuel form the cargo tank through the filtration system, flow meter, and internal piping and then back into the cargo tank. Recirculation shall not use the ground servicing hose. The fuel flow rate shall be controlled by the throttle controller. The control system shall allow recirculation if the primary tank-level sensor (HI) is wet, but not if the secondary tank-level sensor (HI-HI) is wet.

2.7.5 Bottom loading mode. In bottom loading mode, the refueling vehicle shall be capable of filling the cargo tank from a pressurized fuel source through a single point receptacle at flow rates up to 750 gpm with the truck engine off.

2.7.6 Overwing fuel servicing mode. When operating in the overwing fuel servicing mode, the throttle controller and overwing nozzle to be used to control both flow rate and nozzle pressure until the primary pressure control system limits the pressure to no more than 30 ±5 psi. When operating in this mode, the pumping system shall be capable of delivering fuel at rates up to 55 gpm.

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2.8 Mobility. The fully equipped Refueling Vehicle loaded to its gross vehicle weight rating (GVWR) shall:

a. Maximum speed - Maintain a speed of 65 miles per hour (mph) on a dry level paved road for at least 1-mile.

b. Gravel roads - Traverse a graded gravel road (see 6.3.8) at an average speed of 25 miles per hour (mph).

c. Belgian-block or cobblestone roads - Traverse a Belgian-block (see 6.3.2) or cobblestone (see 6.3.3) road at a speed of 20 miles per hour (mph).

d. Cross-country terrain - Traverse cross-country terrain (see 6.3.6) at an average speed of 20 miles per hour (mph).

e. Clearances - Have an angle of approach of at least 15-degrees, an angle of departure of at least 15-degrees, and a ramp breakover angle of at least 10-degrees. Definitions of clearance terms shall be in accordance with SAE J1100.

f. Gradeability - Ascend and descend a 20-degree slope without spillage of any fluid.

g. Ramp negotiation - Negotiate, in each direction, a ramp with a 10-degree minimum slope connecting two horizontal surfaces. The ramp length shall be at least 1.2 times the wheelbase of the Refueling Vehicle. The Refueling Vehicle shall be capable of being driven forward or backward from one horizontal surface across the ramp crest, ascend or descent the ramp, cross the ramp crest, and proceed on the other horizontal surface without any portion of the Refueling Vehicle, other than its tires, contacting any of the three surfaces.

h. Side slope negotiation - Negotiate an 11.5-degree side slope in each direction (right and left side facing up the slope) without spillage of any fluid.

i. Turning diameter - Have a wall-to-wall turning diameter, as defined by SAE J695, of 99 feet maximum in both directions.

j. Steering force - The force required to turn the wheels from lock to lock, with the Refueling Vehicle loaded to its GVWR and the engine at low idle speed, shall not exceed 15-pounds when measured at the steering wheel rim.

k. Parking brakes - Hold the Refueling Vehicle motionless on a 20-degree longitudinal grade in both upgrade and downgrade directions with the transmission in the neutral position. The grade surface shall be dry, hard, and free from loose material. The force required to set the brakes under these conditions shall not exceed 55-pounds.

l. Service brakes - Be in accordance with the performance requirements of Federal Motor Vehicle Safety Standard (FMVSS) Number 121 for air brakes.

2.8.1 Transmission. The Refueling Vehicle shall be equipped with an automatic transmission which shall include a hydraulic torque converter and not less than five forward gear ratios. The net torque capacity and the net power rating of the transmission shall exceed the output ratings of the engine.

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2.8.1.1 Power take-off (PTO). The Refueling Vehicle transmission shall drive the pumping system through the use of a PTO of a split shaft gear box design.

2.8.2 Axles. The Refueling Vehicle shall be equipped with axles rated at least equal to the load imposed on each axle, measured at the ground, with the Refueling Vehicle loaded to its GVWR.

2.8.3 Suspension. The Refueling Vehicle shall be equipped with a suspension system with components having a rated capacity at least equal to the load imposed on each member, measured at the ground, with the Refueling Vehicle loaded to its GVWR. A rear air suspension shall be provided.

2.8.4 Steering. The Refueling Vehicle shall be equipped with a power assisted steering system, with manual steering available in the event of power assist failure. The Refueling Vehicle shall be easy to steer at low speeds (5-25 mph) and provide a controlled steering experience at cruising speeds.

2.8.5 Brakes. The Refueling Vehicle shall be equipped with full-air brakes, with an antilock braking system (ABS), that are in accordance with FMVSS Number 121. Brake linings shall be of non-asbestos material. The braking system, complete with all necessary components, shall include:

a. Air compressor, unloader-head type, engine driven and engine lubricated, air or water cooled, and having a capacity of not less than 340 L/min (12 cubic feet per minute (cfm)).

b. Air storage reservoir(s), each tank equipped with drain, and with safety and check valves between the compressor and the last reservoir tank.

c. Foot control suspended or treadle type.

d. Air control valves.

e. Air pressure gage, visible to the driver.

f. Low air pressure warning, visible and audible.

g. Service brake stop lamp switch.

h. Automatic moisture ejector on air storage reservoir.

i. Automatic slack adjusters on cam type brakes or internal self-adjusting brakes on wedge and disc type brakes on all axles.

j. Brake dust shields on rear.

k. Spring set parking brake.

An air dryer with a replaceable spin on/off desiccant cartridge shall be installed in the air brake system. The dryer shall have the capability of removing not less than 95 percent of the moisture in the air being dried. The dryer shall have a pre-cooler and a filter to screen out oil and solid contaminants. The dryer shall have an automatic self-cleaning cycle and a thermostatically controlled heater to prevent icing of the purge valve. The air dryer shall be in accordance with Technology and Maintenance Council (TMC) RP 637.

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2.8.5.1 Automatic moisture ejector. In addition to the air dryer specified in 2.8.5, the brake system shall also have an automatic moisture ejector installed on the primary air storage reservoir. The automatic moisture ejector shall be plumbed so that a lanyard style air tank drain valve can also be installed. The automatic moisture ejector shall be a Bendix DV-2 or equivalent valve. A heated version of this ejector will not be required.

2.8.6 Tires and wheels. The Refueling Vehicle shall be equipped with single front and rear dual tubeless steel belted radial tires with mud/snow terrain type tread mounted on steel disc wheel assemblies. Tire and wheel assemblies shall be identical at all wheel positions. Tires and wheels shall be in accordance with the *Tire and Rim Association Year Book* requirements for this application. Under no circumstance shall the vehicle maximum governed speed exceed the speed rating of any of the furnished tires. The tires and wheels shall be commercially available.

2.8.6.1 Spare tire and wheel assembly. A spare tire and wheel assembly (the tire mounted on the wheel) shall be provided with the Refueling Vehicle; the assembly shall be identical to the others provided. The spare tire and wheel assembly shall have provisions to secure the assembly during shipping but does not have to have a permanent mount on the vehicle.

2.9 Engine and related equipment. The Refueling Vehicle shall be equipped with Environmental Protection Agency Tier III diesel engine. This requires a National Security Exemption (NSE) from the Clean Air Act. The engine shall be certified to comply with the most stringent Environmental Protection Agency (EPA) non-road diesel engine emission requirements possible while meeting the requirement to operate on fuels containing over 15 parts per million (ppm) sulfur.

2.9.1 Exhaust system. The exhaust system shall be constructed of stainless steel. The muffler(s) shall be constructed of aluminized steel or stainless steel. Exhaust system outlet(s) shall be directed away from personnel accessing any control panel or equipment compartment and the engine air intake. The exhaust system shall meet all applicable requirements of the most recent edition and revision of NFPA 407.

2.9.2 Engine fuels and fuel system.

2.9.2.1 Engine primary fuels. The following shall be primary fuels (see 6.3.9):

a. JP-5, in accordance with MIL-DTL-5624, -40 to 125 °F ambient air temperature.

b. JP-8 in accordance with MIL-DTL-83133 -40 to 125 °F ambient air temperature.

c. F-24, in accordance with NATO AFLP-3747, -40 to 125 °F ambient air temperature.

2.9.2.2 Engine alternate fuels. The following shall be alternate fuels (see 6.3.1):

a. 1-D S15, 1-D S500, 1-D S5000 in accordance with A-A-52557 superseded by ASTM D975, below 60 °F ambient air temperature.

b. 2-D S15, 2-D S500, 2-D S5000 in accordance with A-A-52557 superseded by ASTM D975, above 32 °F ambient air temperature.

c. TS-1 in accordance with GOST 10227-86 with JP-8 additives -40 to 125 °F ambient air temperature.

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2.9.2.3 Engine emergency fuels. The following shall be emergency fuels (see 6.3.7): Jet A, Jet A1 in accordance with ASTM D1655-05, all ambient temperatures.

Note: Jet A and Jet A1 may contain corrosion inhibitor/lubricity improver and icing inhibitors which shall result in properties equal to JP-8.

2.9.3 Engine operator instruments. The following instruments shall be provided to the engine operator:

a. Tachometer.

b. Coolant temperature gauge.

c. Low coolant level indicator.

d. Oil pressure gauge.

e. Oil temperature gauge.

f. Fuel level gauge.

g. Hour meter.

2.10 Electrical system. The Refueling Vehicle shall have a 12-volt, negative ground electrical system in accordance with 3.13.1.2 of SAE ARP1247 except as otherwise specified herein. All electrical circuits shall be protected by circuit breakers located inside the cab. The circuit breakers shall be located in a location that is readily accessible to the operator.

2.10.1 Batteries and battery compartment.

2.10.1.1 Batteries. Batteries shall be of the commercial (see 6.3.4) maintenance-free sealed lead acid, starved electrolyte, gas recombination, spiral wrapped, absorbent gas mat (AGM), top post type in accordance with MIL-B-18013/1.

2.10.1.2 Battery compartment. The batteries shall be enclosed in a corrosion-resistant, weatherproof box or compartment and shall be readily accessible.

2.10.1.3 Battery cables. The battery cables shall be sized to handle the system voltage and current levels, be clearly identified with "+" and "-" or red and black markings and shall not be spliced.

2.10.1.4 Jump Start Connection. The Refueling Vehicle shall be equipped with an external power receptacle for an SAE-AS90347 type plug to facilitate jump starting operations.

2.10.2 Winterization system. A winterization system shall be provided for starting in temperatures below 0 °F. The winterization system may include heaters for engine coolant, engine oil, and the fuel tank, as well as battery warmers. The winterization system shall be designed to operate from both an external 120/208 volt AC, 60 Hz power source utilizing the external electrical connections and an external 220/308 volt AC, 50 Hz power source utilizing external connections. The winterization system shall incorporate high-temperature shutoff switches to prevent overheating of any fluid or component. A labeled indicator light shall be installed on the instrument panel to indicate when the vehicle is connected to external power.

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2.10.3 Plugs and connectors. Plugs and connectors shall be in accordance with NEMA WD 6.

2.11 Technical data.

2.11.1 Manufacturer publications. The contractor shall provide the following manufacturer's publications in Indexed Portable Document Format (IPDF). One digital copy on disc shall be furnished to AFLCMC/WNZ 30 days prior to the first hardware delivery. All manuals shall be delivered with either unlimited data rights or as a minimum, a Government Purpose License, to allow the Air Force to electronically store, print, distribute and update the manual. Manuals are subject to government review and approval in accordance with MIL-PRF-32216. Any changes or corrections noted by the Air Force shall be corrected and updated pages or manuals shall be submitted. Only one digital copy is required per order regardless of quantity of units.

2.11.1.1 Operator's manual. The operator's manual shall include all information required for the safe and efficient operation of the vehicle. The operator's manual shall include at least the following:

a. General description and step-by-step instructions for the operation of the vehicle and its auxiliary equipment.

b. Checklists, lubrication charts, and basic troubleshooting procedures for the daily maintenance inspection and mission readiness checks that the operator is expected to perform.

c. Schedules for required preventative maintenance and required periodic maintenance.

d. Full descriptions and illustrations for the location and function of all controls.

e. List of all checks and adjustments needed in preparation for placing the vehicle into service upon receipt from the contractor.

f. Preparation Instructions for shipment or storage.

g. Warranty information and a list that details the period of the warranty for the complete vehicle and for any component that exceeds the warranty of the complete vehicle. Addresses and telephone numbers shall be provided for all warranty providers listed.

2.11.1.2 Service and repair manuals. Service and repair manuals shall be provided for the chassis, engine, transmission, tanker body, and equipment. The service manuals shall identify any special tools and test equipment required and shall cover troubleshooting and maintenance as well as minor and major repair procedures. The text shall contain performance specifications, tolerances, fluid capacities, current data, voltage data, resistance data, test procedures, and any illustrations or exploded views that may be required to perform proper maintenance by qualified mechanics. The service and repair manuals shall contain an alphabetical subject index as well as a table of contents. The manual shall contain at least the following where applicable:

a. Pneumatic, electrical, and fuel schematics.

b. Schedules for required preventative maintenance and required periodic maintenance.

c. Troubleshooting, servicing, disassembly, repair, and assembly procedures.

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2.11.1.3 Illustrated parts breakdown. An illustrated parts breakdown (IPB) manual shall be provided for the chassis, engine, transmission, tanker body, and equipment. IPB manuals shall include illustrations and exploded views as needed to properly identify all parts, assemblies, subassemblies, and special equipment. All components of assemblies shown in illustrations or exploded views shall be identified by reference numbers that correspond to the reference numbers in the parts list. All purchased parts shall be cross-referenced with the original manufacturer's name and part number. The IPB manual shall provide the description and quantity of each item used per vehicle. The size, thread dimensions, torque specifications, and special characteristics shall be provided for all nonstandard nuts, bolts, screws, washers, grease fittings, and similar items. The IPB manuals shall contain a numerical index. The manuals shall contain a list of all the component vendor names, addresses, and telephone numbers referenced in the parts list.

2.11.1.4 Mailing address. The electronic copy of the manufacturer's publications shall be sent to the following address:

If sending via USPS:
Support Equipment & Vehicles Division
Refueler IPT
AFLCMC/WN-SEV
Attn: Mr. Brandon Hammontree
235 Byron Street, Suite 19A
Robins AFB, GA 31098-1813

If sending via Fed Ex of UPS:
Support Equipment & Vehicles Division
Refueler IPT
AFLCMC/WN-SEV
Attn: Mr. Brandon Hammontree
235 Byron Street, Suite 19A
Bldg. 300WW South End

2.11.2 Service bulletins. The contractor shall prepare service bulletins as required to notify the government of new maintenance problems and/or procedures to be implemented on the contractor's equipment prior to the issuance of permanent changes to the technical manuals. Service bulletins shall also be prepared when the program office has identified a trend of a potential problem or failure that has occurred at field level. The service bulletins shall identify any potential safety hazards and shall be submitted in a Microsoft Office digital format.

2.12 Workmanship. The Refueling Vehicle, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Workmanship objectives shall include freedom from blemishes, defects, burrs and sharp corners and edges; accuracy of dimensions, surface finish, and radii of fillets; thoroughness of welding, painting, and riveting; marking of parts and assemblies; and proper alignment of parts and tightness of assembly fasteners.

2.12.1 Bolted connections. Bolt holes shall be accurately punched or drilled and shall be deburred. Threaded fasteners shall be tight and shall not work loose during testing or service usage.

2.12.2 Riveted connections. Rivet holes shall be accurately punched or drilled and shall be deburred. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the component.

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2.12.3 Gear and lever assemblies. Gear and lever assemblies shall be properly aligned and meshed and shall be operable without interference, tight spots, loose spots, or other irregularities. Where required for accurate adjustment, gear assemblies shall be free of excessive backlash.

2.12.4 Cleaning. The Refueling Vehicle shall be thoroughly cleaned. Loose, spattered, or excess solder; welding slag; stray bolts, nuts, and washers; rust; metal particles; pipe compound; and other foreign matter shall be removed during and after final assembly.

3. REGULATORY REQUIREMENTS.

3.1 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with 23.403 of the Federal Acquisition Regulation (FAR). However, used, rebuilt, or refurbished items shall not be provided.

3.2 Green Procurement Program. Green Procurement Program (GPP) is a mandatory federal acquisition program that focuses on the purchase and use of environmentally preferable products and services. GPP requirements apply to all acquisitions using appropriated funds, including services and new requirements. FAR 23.404(b) applies and states the GPP requires 100% of EPA designated product purchase that are included in the Comprehensive Procurement Guidelines list that contains recovered materials, unless the item cannot be acquired: a) competitively within a reasonable timeframe; b) meet appropriate performance standards, or c) at a reasonable price. The prime contractor is responsible for ensuring that all subcontractors comply with this requirement.

4. PRODUCT CONFORMANCE PROVISIONS.

The products provided shall meet the salient characteristics of this Product Description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace, modified as necessary to comply with the requirements herein. The Government reserves the right to require proof of such conformance.

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First production inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First production inspection. The first production Refueling Vehicle shall be subjected to the analyses, demonstrations, examinations, and tests described in 4.5.1 through 4.5.8. The contractor shall provide or arrange for all test equipment and facilities. Except as otherwise specified, all testing in which the engine is operated shall be performed using JP-8 turbine fuel.

4.3 Conformance inspection. Each production Refueling Vehicle shall be subjected to the examination described in 4.5.1.

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4.4 Inspection requirements.

4.4.1 General inspection requirements. Apparatus used in conjunction with the inspections specified herein shall be laboratory precision type, calibrated at proper intervals to ensure laboratory accuracy.

4.4.2 Data. During all testing specified herein, at least the following data, unless not applicable, shall be recorded at intervals not to exceed 30 minutes. Additional data or shorter intervals shall be provided as appropriate for any specific test.

a. Date.

b. Time started.

c. Time finished.

d. Ambient temperature.

e. Ambient humidity.

4.4.3 Test rejection criteria. Throughout all tests specified herein, the Refueling Vehicle shall be closely observed for the following conditions, which shall be cause for rejection.

a. Failure to conform to design or performance requirements specified herein or in the contractor's technical proposal.

b. Any spillage or leakage of any liquid, including fuel, coolant, lubricant, or hydraulic fluid, under any condition, except as allowed herein.

c. Thermal or structural failure of any component, including permanent deformation, or evidence of impending failure.

d. Evidence of excessive wear. If excessive wear is suspected, the original equipment manufacturers (OEM's) specifications or tolerances shall be utilized for making a determination.

e. Evidence of corrosion or deterioration.

f. Misalignment of components.

g. Conditions that present a safety hazard to personnel during operation, servicing, or maintenance.

h. Interference between the Refueling Vehicle components or between the Refueling Vehicle, the ground, and all required obstacles, with the exception of normal contact by the tires.

i. Evidence of undesirable mobility characteristics, including instability in handling during cornering, braking, and while traversing all required terrain.

j. Shutdown faults from:

1) Engine cooling system.

2) Engine lubrication system.

3) Engine protective circuits.

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4.5 Test methods.

4.5.1 Examination of product. Each Refueling Vehicle shall be examined to verify compliance with the requirements herein prior to accomplishing any other demonstrations or tests listed in 4.5. A contractor-generated, Government-approved checklist (part of the test procedure) shall be used to identify each requirement not verified by an analysis, certification, demonstration, or test, and shall be used to document the examination results. Particular attention shall be given to materials, workmanship, dimensions, surface finishes, protective coatings and sealants and their application, welding, fastening, and markings. Proper operation of each Refueling Vehicle function shall be verified. Certifications and analyses shall be provided in accordance with Table III. Each production Refueling Vehicle shall be inspected to a Government-approved reduced version of the checklist.

TABLE III. Certifications and analyses.

Paragraph	Required Certifications and Analyses
2.2.4.5 Noise test.	Contractor certification that the noise specifications are in accordance with the requirements set forth in 2.2.4.5.
2.2.5 Electromagnetic interference (EMI).	Contractor certification that EMI specifications are in accordance with the requirements set forth in 2.2.5.
2.3.1 Operating temperature range.	Contractor certification that operating temperature specifications are in accordance with the requirements set forth in 2.3.1.
2.3.2 Storage temperature range.	Contractor certification that storage temperature specifications are in accordance with the requirements set forth in 2.3.2.
2.3.3.1 Rain.	Contractor certification that rain specifications meet the requirements set forth in 2.3.3.1.
2.3.3.2 Snow.	Contractor certification that the snow specifications meet the requirements set forth in 2.3.3.2.
2.3.3.5 Solar radiation.	Contractor certification that the Refueling Vehicle performance is not adversely affected by full time exposure to solar radiation, such as those conditions encountered in desert environments.
2.3.3.6 Fungus.	Contractor certification that the materials used in construction of the Refueling Vehicle are fungus resistant or suitably treated to resist fungus.

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TABLE III. Certifications and analyses - Continued.

Paragraph	Required Certifications and Analyses
2.5.1 Surface transportability.	Contractor surface transportability analysis (see 4.5.6.1.1) and certification that the Refueling Vehicle is transportable via all modes of surface shipment (highway, rail, and water) in accordance with MIL-STD-1366 and shall be capable of withstanding the mechanical shock and vibration characteristics of highway, rail, and water transport.
2.5.2 Air transportability., and 2.5.2.1 Shoring.	Contractor air transportability analysis (see 4.5.6.2).
2.5.3 Tie downs.	Contractor tie down provision analysis (see 4.5.6.4.1).
2.8 Mobility.	Contractor certification that the mobility specifications will meet the requirements set forth in 2.8.
2.9 Engine and related equipment., 2.9.2.1 Engine primary fuels., 2.9.2.2 Engine alternate fuels., and 2.9.2.3 Engine emergency fuels.	Engine manufacturer certification that the engine is in accordance with all applicable requirements, including exhaust emissions standards and fuels.

4.5.2 Noise test. Certification for noise testing shall be acceptable in lieu of a first article test (see Table III).

4.5.3 Electromagnetic interference test. Certification for EMI testing shall be acceptable in lieu of a first article test (see Table III).

4.5.4 Environmental testing.

4.5.4.1 High temperature storage and operation test. Certification for high temperature and operation testing shall be acceptable in lieu of a first article test (see Table III).

4.5.4.2 Low temperature storage and operation test. Certification for low temperature and operation testing shall be acceptable in lieu of a first article test (see Table III).

4.5.4.3 Precipitation.

4.5.4.3.1 Rain test. Certification for rain testing shall be acceptable in lieu of a first article test (see Table III).

4.5.4.3.2 Snow load test. Certification for snow testing shall be acceptable in lieu of a first article test (see Table III).

4.5.4.3.3 Ice accretion test. Certification for ice accretion testing shall be acceptable in lieu of a first article test (see Table III).

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4.5.4.4 Salt fog test. Certification for salt fog testing shall be acceptable in lieu of a first article test (see Table III).

4.5.4.5 Sand and dust test. Certification for sand and dust testing shall be acceptable in lieu of a first article test (see Table III).

4.5.5 Weight and dimension tests.

4.5.5.1 Weight and center of gravity test. The weight, center of gravity, and axle weights of a first production Refueling Vehicle shall be measured to demonstrate compliance with the weight requirement of 2.4 and the axle weight requirement of 2.5.2.2.

4.5.5.2 Dimension measurement. A first production Refueling Vehicle shall be measured to demonstrate compliance with the dimensional requirements of 2.4.

4.5.6 Transportability verification.

4.5.6.1 Surface transportability verification.

4.5.6.1.1 Surface transportability analysis. An engineering analysis shall be performed to demonstrate compliance with 2.5.1. The engineering analysis shall utilize the data for road transportation in accordance with MIL-STD-810, Method 514.7, Table 514.7C-II.

4.5.6.2 Air transportability analysis. An engineering analysis shall be performed to demonstrate compliance with the air transportability requirements of 2.5.2, 2.5.2.1, and 2.5.2.4. The analysis shall include the tie downs and all major components and their ability to withstand the accelerations specified in 2.5.2. The evaluation shall also include a dimensional analysis for the Refueling Vehicle while traversing the ramp and while loaded aboard C-130, C-17, and C-5 aircraft.

4.5.6.3 Equipment removal and reconfiguration demonstration. A first production Refueling Vehicle shall be configured for transport on C-130, C-17, and C-5 aircraft and then reconfigured for operation to demonstrate compliance with 2.5.2.4. It shall be demonstrated that the forces required do not exceed those allowed in MIL-STD-1472.

4.5.6.4 Tie down provision verification.

4.5.6.4.1 Tie down provision analysis. An engineering analysis shall be performed to demonstrate compliance with the tie down provision requirements of 2.5.3.

4.5.6.4.2 Tie down provision test. A first production Refueling Vehicle shall be tested to demonstrate compliance with the tie down provision requirements of 2.5.3.

4.5.7 Maintainability demonstration. All recommended preventive maintenance tasks shall be performed, and the task times shall be recorded. It shall be demonstrated that the forces required do not exceed those allowed in MIL-STD-1472.

4.5.8 Mobility tests. Certification for mobility testing shall be acceptable in lieu of a first article test (see Table III).

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5. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

6. NOTES.

6.1 Sources of documents.

6.1.1 Government Documents. Military Specifications, Standards, and Handbooks referenced herein may be obtained from https://quicksearch.dla.mil or the Standardization Documents Order Desk, 700 Robbins Ave, Bldg 4, Section D, Philadelphia, PA 19111-5094.

6.1.2 Air Force Instructions (AFI). Copies of the AFI may be obtained from https://www.e-publishing.af.mil.

6.1.3 Code of Federal regulations (CFR). Copies of the CFR are available online at https://www.ecfr.gov.

6.1.4 Federal Acquisition Regulation (FAR). Copies of the FAR may be obtained from https://www.acquisition.gov.

6.1.5 Advancing Standards Transforming Markets (ASTM). Copies of the ASTM may be obtained from https://www.astm.org.

6.1.6 American Welding Society (AWS). Copies of the AWS may be obtained from https://www.aws.org.

6.1.7 National Electrical Manufacturers Association (NEMA). Copies of the NEMA may be obtained from https://www.makeitelectric.org.

6.1.8 National Fire Protection Association (NFPA). Copies of the NFPA may be obtained from https://www.nfpa.org.

6.1.9 Society of Automotive Engineers (SAE). Copies of the SAE may be obtained from https://www.sae.org.

6.1.10 Technology and Maintenance Council (TMC). Copies of the TMC may be obtained from https://www.trucking.org.

6.1.11 Tire and Rim Association documents. Application for copies should be addressed to The Tire and Rim Association, Inc., 175 Montrose West Ave., Suite 150, Copley OH 44321, https://www.us-tra.org.

6.1.12 RussianGost (GOST). Copies of the GOST may be obtained from https://www.russiangost.com.

6.2 Ordering data. Acquisition documents should specify the following:

a. Finish color required (Semi-gloss green or tan (see 2.2.1.3)).

b. Base location for jet fuel marking (see 2.2.6).

c. If extreme climatic hoses are required (see 2.2.13.1).

6.3 Definitions.

6.3.1 Alternate fuel. A fuel that can be used in place of the primary fuel with a possible loss of efficiency, engine adjustments may be necessary. The use of an alternate fuel may result in a change of maintenance or overhaul cost.

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6.3.2 Belgian-block. A Belgian-block course consists of a very rough, undulating, hard surface unaffected by changes in climatic conditions. The standard is found at Aberdeen Proving Grounds, Maryland.

6.3.3 Cobblestone. A naturally rounded stone larger than a pebble and smaller than a boulder; *especially*: such a stone used in paving a street or in construction.

6.3.4 Commercial item.

a. Any item, other than real property, that is of a type customarily used by the general public or by non-Governmental entities for purposes other than Governmental purposes, and:

1) Has been sold, leased, or licensed to the general public; or

2) Has been offered for sale, lease, or license to the general public;

b. Any item that evolved from an item described in paragraph (1) of this definition through advances in technology or performance and that is not yet available in the commercial marketplace, but will be available in the commercial marketplace in time to satisfy the delivery requirements under a Government solicitation;

c. Any item that would satisfy a criterion expressed in paragraphs (1) or (2) of this definition, but for:

1) Modifications of a type customarily available in the commercial marketplace; or

2) Minor modifications of a type not customarily available in the commercial marketplace made to meet Federal Government requirements. Minor modifications means modifications that do not significantly alter the non-Governmental function or essential physical characteristics of an item or component or change the purpose of a process. Factors to be considered in determining whether a modification is minor include the value and size of the modification and the comparative value and size of the final product. Dollar values and percentages may be used as guideposts, but are not conclusive evidence that a modification is minor;

d. Any combination of items meeting the requirements of paragraphs (1), (2), (3), or (5) of this definition that are of a type customarily combined and sold in combination to the general public;

e. Installation services, maintenance services, repair services, training services, and other services if:

1) Such services are procured for support of an item referred to in paragraph (1), (2), (3), or (4) of this definition, regardless of whether such services are provided by the same source or at the same time as the item; and

2) The source of such services provides similar services contemporaneously to the general public under terms and conditions similar to those offered to the Federal Government;

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f. Services of a type offered and sold competitively in substantial quantities in the commercial marketplace based on established catalog or market prices for specific tasks performed or specific outcomes to be achieved and under standard commercial terms and conditions. For purposes of these services:

1) "Catalog price" means a price included in a catalog, price list, schedule, or other form that is regularly maintained by the manufacturer or vendor, is either published or otherwise available for inspection by customers, and states prices at which sales are currently, or were last, made to a significant number of buyers constituting the general public; and

2) "Market prices" means current prices that are established in the course of ordinary trade between buyers and sellers free to bargain and that can be substantiated through competition or from sources independent of the offerors.

g. Any item, combination of items, or service referred to in paragraphs (1) through (6) of this definition, notwithstanding the fact that the item, combination of items, or service is transferred between or among separate divisions, subsidiaries, or affiliates of a contractor; or

h. A non-developmental item, if the procuring agency determines the item was developed exclusively at private expense and sold in substantial quantities, on a competitive basis, to multiple State and local Government. (Reference the Federal Acquisition Regulation (FAR) 2.101).

6.3.5 Common hand tool. A non-powered tool that is likely to be found in a typical mechanic's toolbox. Common hand tools include open end, boxed end, combination, socket (both 6- and 12-point in both standard and deep-well), and hex key wrenches, in SAE sizes up to and including 1-inch and metric sizes up to and including 25mm; ratchet handles, extensions, and swivels; slotted and Phillips-head screwdrivers; regular and snap-ring pliers; and a ball-peen hammer.

6.3.6 Cross-country terrain. Cross-country terrain consists of unimproved open fields, broken ground, loose sand, and gravel road encountered by tactical vehicles supporting an army in the field.

6.3.7 Emergency fuel. A fuel that can be used when the primary and alternate fuels are not available. This fuel is not for use on a continuing basis but is to be only when the primary and alternate fuel and equipment fuels are unobtainable and operation is mission essential. Engine life could be severely impacted.

6.3.8 Gravel road. A gravel road is a level or rolling gravel trail.

6.3.9 Primary fuel. The fuel or fuels used during tests to demonstrate system performance (contract compliance). Substitutions are presented as Alternate and Emergency fuels.

6.3.10 Special tool. A tool that is not commercially and readily available from a source other than the Refueling Vehicle contractor.

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6.4 Key words.

Refueler
Jet Fuel
Batteries
3000 Gallon
Refueling Vehicle

MILITARY INTEREST

Custodian:
Air Force - 184

Preparing activity:
Air Force - 184

(Project 1730-2026-002)

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